

# **Women's economic autonomy in the Dutch Republic and maritime-induced male absence**

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## **Abstract**

This thesis studies the causal effect of maritime-induced male absence on women's economic autonomy in the early-modern Dutch Republic, proxied respectively by Dutch East India Company (VOC) seafarer departures and the literacy gender gap. To fit the conceptual mechanism that maritime-induced male absence caused women to acquire the human capital necessary to survive in society, I exploit city-level variation in VOC seafarer departures and literacy gender gaps. Using a continuous difference-in-differences design, I find that a one standard deviation increase in VOC seafarer departures relative to the baseline population in 1600 leads to a 2.84 percentage point decrease in the literacy gender gap when controlling for longitude and latitude. The results are highly robust to population growth controls and less robust to a leave-one-out test. The findings contribute to literature on the persistent effects of historical shocks and have implications for normative debates on colonialism.

**Keywords:** literacy gender gap, maritime male absence, Dutch Republic, Dutch East India Company (VOC), historical natural experiment

**Word count:** 7393

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# 1 Introduction

The rise of the Dutch Republic in the 17th century marked a large increase in the relative importance of the maritime trading sector, where long and high-mortality voyages carried out by the Dutch East India Company (VOC) were necessary to both fund the independence war against the Spanish Empire and contest Spanish and Portuguese dominance in these areas, later creating the basis for the Dutch Golden Age (de Vries & van der Woude, 1997; Israel, 1995). This independence war, the Eighty Years' War, was fought also along religious lines: the powder keg of encroaching centralisation and high taxes around the northern Dutch cities erupted due to the unfair and violent Spanish treatment of Protestants and the cities and their elites that were perceived to be too lenient (Israel, 1995, pp. 129–160).

This presents an interesting historical natural experiment (Cantoni & Yuchtman, 2020) to study from a gendered perspective. While the general effects on the Dutch Republic's economic prosperity are well known, little is known about the marginalised groups at home, and along gender lines, the group of interest is Dutch women. As a shock to the adult sex ratio, maritime-induced male absence can have significant effects on women's economic outcomes. Thus, I explore the following research question: *How did prolonged and permanent male absence resulting from voyages of the Dutch East India Company affect women's economic autonomy in the early-modern Dutch Republic?*

This thesis provides supportive evidence for the hypothesis that women's economic autonomy in the Dutch Republic was positively affected by maritime-induced male absence. Through a continuous difference-in-differences design, I exploit variation in the literacy gender gap and seafarer departures across Dutch cities as measures for respectively women's economic autonomy and maritime-induced male absence. This analysis shows a significant closing effect of VOC seafarer departures on the literacy gender gap in 1750, which is almost entirely driven by increases in female literacy.

The structure is as follows: Section 2 provides an overview of the literature on historical shocks to the adult sex ratio and increases in religious competition, which identifies the research gap through the review. Section 3 describes the conceptual framework behind the hypothesis of the research question and outlines data sources, data modifications and summary statistics. Section 4 discusses the empirical strategy and presents the results of the difference-in-differences analysis. Finally, Section 5 discusses the results, their interdisciplinary connections and societal implications and general conclusions.

## **2 Literature review**

A vast strand of economic literature quantitatively explores historical shocks to make causal claims about the resulting short- and long-run outcomes (Cantoni & Yuchtman, 2020). Within this strand, the presence and strength of numerous causal channels, like wartime shocks resulting in male absence, have been extensively studied in the context of their influence on women's outcomes (Acemoglu et al., 2004; Boehnke & Gay, 2022; Goldin, 1991; Goldin & Olivetti, 2013). This review thus examines the effects of shocks to the adult sex ratio and increases in religious competition, before identifying research gaps guided by qualitative historian work on women in the early-modern Dutch Republic. In turn, this review then identifies several research gaps regarding the underexplored effects of peacetime male absence on women's outcomes and the lack of quantitative work on women in the Dutch Republic.

### **2.1 Shocks to the adult sex ratio**

Historical shocks to the adult sex ratio can have profound impact on the roles of women in society, where pre-existing cultural norms and political institutions restricting women's economic emancipation were, at minimum, temporarily overridden out of necessity. Goldin (1991) first quantitatively explored the broad topic of wartime effects on women's employment. Raising doubts around the notion dominant under historians that the wartime absence of husbands during World War II was a critical juncture afterwards for women's employment in the United States, Goldin (1991) found that while the wartime absence of husbands in the short-run did induce women to enter the labour force, little evidence exists for the long-run wartime effects on women's employment.

Later, Acemoglu et al. (2004) provided more extensive evidence using cross-state variation in wartime male mobilisation, which, albeit not the main purpose of their work, reinforced Goldin's (1991) short-run findings and showed, through a difference-in-differences logic, a somewhat persisting post-war effect on female employment in states with higher mobilisation rates in 1950. Similarly, Goldin and Olivetti (2013) used mobilisation data, marriage status and other demographic attributes of women for a triple difference strategy to specifically show that American states with higher mobilisation rates experienced larger short-run positive impacts on female employment. Interestingly, they also revisited Goldin's (1991) lack of long-run findings, showing that heterogeneous positive effects existed for highly educated women who were not driven back out of the labour force after World War II ended. Occupation type also mattered, albeit correlated with educational background: highly educated women were more likely to work in white-collar sectors, which were more welcoming to female workers than manufacturing sectors. Thus, this collection of literature over time shows that

wartime shocks especially matter for short-run boosts in aggregate female employment, while their long-run effects are only discernible when accounting for heterogeneity in women's educational backgrounds.

Adding onto this, more recent work by Boehnke and Gay (2022) approached this topic with a novel approach using county-level French wartime fatalities in World War I, showing how higher military fatalities positively affected post-war female labour force participation. The authors attribute this effect to supply-side factors, where missing men for unmarried women and widowed women were pushed by their circumstances to persistently participate in the labour force. Thus, statistics such as wartime fatalities, which directly and lastingly affect the adult sex ratio more than mobilisation rates, provide more convincing evidence that long-run effects of wartime shocks exist when directly examining variation in wartime mortality. Altogether, wartime shocks to the adult sex ratio can have drastic impact on women's lives, where material circumstances force widowed and single women into the labour force and cultural participation barriers can be temporarily and sometimes even permanently diminished.

Moving on, persistent colonial violence can also function as a shock to the adult sex ratio and lead to more longer-run effects compared to wartime shocks. Compared to a sharp and short wartime shock, colonial legacies such as the presence of the slave trade led over time to African regions with female-biased sex ratios. Parallel to the findings of previously discussed literature, this led to women temporarily taking over male labour tasks. In addition to reinforcing the established findings, the author also found long-run cultural effects where women whose ancestors were exposed to the slave trade are now more likely to be participating in the labour force (Teso, 2019).

The length of this shock in conjunction with the long-run cultural effects, therefore, suggests that long-lasting shocks compared to short wartime shocks can lead to persistent shifts in cultural norms and gender roles. The basis of these measured long-run cultural effects is further reinforced by the related and recognised work of Alesina et al. (2013) on the presence of the tradition of plough agriculture on contemporary gender roles, where its resulting gendered division of labour persistently shaped unequal attitudes about the roles of men and women. Thus, the nature of a historical shock may matter for its long-run effects in particular: the effects of far longer and more persistent male absence could also persist in the intergenerational transmission of affected cultural norms and gender roles.

## **2.2 Increases in religious competition: the Protestant Reformation and the Brethren of the Common Life**

Religious shocks are especially relevant for the early-modern experiences of women, with the most prominent one in Europe being the Protestant Reformation. Protestantism

was conducive towards the economic progress of women from the 16th century onward in Western Europe, as this Christian variant, for instance, promoted the literacy of both women and men as a means to be able to read and interpret the Bible themselves (Becker & Woessmann, 2008), reducing their dependency on the readings of established religious institutions such as the Catholic Church. Becker and Woessmann (2008) find through an instrumental variable approach using county- and town-level data from the 1816 Prussian population census that a larger share of Protestants in a county or town is associated with a smaller literacy gap between genders, suggesting that the presence of the Reformation, in line with its doctrines, promoted gender equality in this specific aspect that forms an essential component of human capital valued by higher-skill occupations.

Shifting closer to the Dutch context of interest, Akçomak et al. (2016) studied the effect of a pre-Protestant yet similar movement that supplemented the teachings of the Catholic Church, rather than contesting its dominant status, namely the Brethren of the Common Life (BCL). Founded in the Dutch city of Deventer in 1374, this movement similarly promoted literacy under all Christian believers to be able to independently read the Bible. Using a similar instrumental variable approach utilised by Becker and Woessmann (2008) where the distance to the Dutch city of Deventer is used as an instrument, Akçomak et al. (2016) note, alongside their primary findings that the historical presence of the BCL was associated with a long-run positive effect on 1600 literacy rates in Dutch cities, that slightly smaller yet still significant effects of the BCL can be observed when only taking female literacy data into account.

Thus, these religious shocks also had profound effects on women's outcomes, where a strand of literature focusing on the introduction of religious competition provides evidence that the human capital accumulation of women was stimulated by the presence of religious movements and doctrines conducive and friendly towards this end. In the early-modern context of the Dutch Republic, the examination of the two discussed causal mechanisms is especially relevant, as these phenomena were simultaneously present in the taken period of study. Qualitative work by historians point to the unbalanced sex ratio when considering why the employment rate of Dutch women was high, where the prominence of the Dutch East India Company in the economy led to both 'the departure of thousands of men of marriageable age' and left-behind married women (Schmidt, 2025, p. 20).

## **2.3 Research gap**

The review of the literature on historical shocks and women's outcomes thus shows that there is still academic contestation over the presence and strength of long-run effects of historical shocks: the length of the shock's presence seems to matter in this

aspect. Additionally, to my knowledge, no quantitative work exists on why the economic autonomy of Dutch women seemed higher than in other countries at the time, where travellers' accounts 'portrayed early modern Dutch women as economically active, entrepreneurial, independent, and dominant' (Schmidt, 2025, p. 17) relative to their foreign countries. Moreover, none of the reviewed literature has studied the effects of prolonged male absence caused by a peacetime context rather than a wartime context.

Thus, the identified research gap is one where firstly, the relative explanatory strength of shocks to the adult sex ratio and shocks to religious competition is still unknown, with none of the reviewed literature exploring historical cases where these two phenomena overlapped. Secondly, the presence of long-run effects such as cultural persistence resulting from long-lasting shocks is underexplored in the context of dominant and developed countries, where the decline of the Dutch East India Company over time also led to less male absence. To summarise, there exists also a lack of literature on whether temporary yet long-lasting shifts in the role of women in the domestic economy outlast the eventual return of men. By examining the prolonged and permanent male absence caused by the voyages of the Dutch East India Company, my thesis aims to address this research gap.

### **3 Conceptual framework and data sources**

To analyse women's economic autonomy in the Dutch Republic, I use the gap between female and male literacy as an observable proxy outcome, which I argue is sufficiently representative of economic autonomy. Next, I utilise seafarer departures as an observable exposure measure to maritime-induced male absence, which serves as an exogenous shock to explain differences in the literacy gender gap. This theorised mechanism aligns with previous literature describing how higher female literacy is associated with improved economic outcomes (Becker & Woessmann, 2008; Goldin, 2006). Moreover, the observation of a skill differential is related to the theoretical literature on women's bargaining power given their threat points in marriage (Manser & Brown, 1980; McElroy & Horney, 1981).

In the following section, I elaborate why the literacy gender gap sufficiently proxies for female economic autonomy and how VOC seafarer departures represent an exogenous shock to the adult sex ratio, subsequently affecting women's economic autonomy. Afterwards, city-level literacy data and VOC employee records are described and operationalised for this conceptual framework. Finally, summary statistics and pre-treatment covariate balance tests are presented alongside the implications of the latter for the following empirical analysis in Section 4.

### **3.1 The literacy gender gap as a measure of economic autonomy**

Literacy rates, divided by gender, are an indicator of human capital accumulation between women and men. The level of human capital accumulation reflects the decisions and social positions of women and men: all individuals would choose to maximise their welfare, but they may face cultural and institutional barriers that dissuade them from becoming literate. In principle, human capital is generally necessary to access higher-skilled occupations, which correspondingly pay higher wages. Inside the historical maritime context of the Dutch Republic, occupations related to maritime trade required workers with higher-level roles to ‘keep records, [and] make toll and tariff declarations’ (de Vries & van der Woude, 1997, p. 172). The ability to read and write was therefore essential for attaining higher-level positions inside the maritime sector, which correspondingly would come with higher wages. This was no different for the Dutch East India Company, where positions that required higher levels of skill such as literacy correspondingly returned higher wages (Rei, 2014). More generally yet also more specifically for women, literacy also aided women in running or taking over businesses of their parent or spouse (van den Heuvel, 2007, pp. 48, 54).

The level of literacy acquired, however, is constrained by previously established male human capital advantages, which create dissuading and segregatory effects. Firstly, women are less inclined to acquire literacy and subsequently end up in lower-skilled occupations if the competition with men on the labour market is unbalanced, given that this unbalanced competition leads to lower returns to literacy for women. Secondly, outside the labour market context, women, given the pre-existing but not restrictive social norms about their domestic role in the household (Schmidt, 2011), may then be more inclined to pursue marriage as a means of subsistence. Studying the literacy gender gap therefore signals the relative economic position of women and men, which thus forms a crude yet sufficient proxy for economic autonomy.

### **3.2 Seafarer departures as an exogenous shock to the adult sex ratio and marriage markets**

The prolonged and permanent absence of seafarers in cities serves as a measure of maritime-induced male absence, exogenously shocking the adult sex ratio: nominally, only men were allowed to sail with the VOC, and as the largest maritime company driving the Dutch Golden Age, the subsequently large number of absent seafarers leads to a substantial shock affecting local marriage markets. As a result, unmarried women face a lower supply of available men and married women are left behind by their husband. This exogenous shock, in turn, also positively spills over to Dutch women’s autonomy: voyages with the VOC were lengthy and deadly (Bruijn et al., 1987), leading to many



married women taking sole charge of the household in cities where the VOC was highly present (van der Heijden & van den Heuvel, 2007).

Despite the existence of a support system where seafarers' wives could receive wage transfers from their husbands, it was nonetheless necessary to search for work to make ends meet. To obtain occupations that offer higher wages, acquiring literacy would have been necessary to avoid low-paying occupations or activities considered immoral and illegal at that time, such as prostitution (van der Heijden & van den Heuvel, 2007).

Married women were also able to obtain legal capacity during the period in which their husband was away for a prolonged period, allowing seafarers' wives to, for example, collect inheritances and purchase property, which they would otherwise not be able to do without legal capacity (van der Heijden & van den Heuvel, 2007). Unmarried women were already relatively more independent both legally and in spirit in the Dutch Republic than in other European countries (de Vries & van der Woude, 1997, pp. 598–600), signalling less restrictive gender norms after the onset of the Republic when it came to women attaining significant roles in the economy and society. All in all, the institutional framework was generally supportive of seafarers' wives.

Thus, VOC seafarer departures can be used as an exogenous shock that can positively affect women's economic autonomy: the shock works through a market mechanism where there are fewer men available for marriage, and the shock compels seafarers' wives to supplement their insufficient income and granting them the legal capacity to manage household affairs in the face of male absence.

### **3.3 Data on literacy rates, VOC employee records and city populations**

Data on female and male literacy rates in the Dutch Republic are provided by Akçomak et al. (2016), who aggregated literacy rates at the city level by differentiating between hand signatures and simple marks of brides and grooms in the Amsterdam Marriage Registers as a proxy for literacy (Hart, 1976). The place of former residence was registered in the records, allowing for literacy rates to be traced back to the cities from which the bride and the groom originated from (Akçomak et al., 2016). This dataset contains 34 cities with female and male literacy rates at 1600, 1675 and 1750, geographically covering both small and large cities across the Dutch Republic.

To first construct a continuous exposure measure to maritime-induced male absence, I use the enriched *VOC-opvarenden* [VOC seafarers] dataset, which is constructed from 774,200 muster records of seafarers in the Dutch East India Company from 1633 to 1794 (Petram et al., 2024). Given the high mortality rate and lengthiness of these voyages (Bruijn et al., 1987), seafarer departures can be used as a measure of prolonged and permanent male absence by their place of origin, aligning with a related strand of literature that uses war mobilisation and fatalities as similar sources of variation to

estimate causal effects on women’s outcomes (Acemoglu et al., 2004; Boehnke & Gay, 2022; Goldin & Olivetti, 2013). Additionally, these records exclusively comprised of male names, showing that these records can serve as a reliable measure of male seafarer departures (Petram et al., 2024). I filter for disambiguated people and whether the place and date of their first departure is registered, reducing the dataset to 459,766 observed individuals. Seafarer departures in the VOC dataset are aggregated into treatment intensity per city from the first recorded year of departure, 1633, to 1750. By making use of Petram et al. (2024)’s standardised place names, handwritten place names are matched by their corresponding standardised place names, which also include information about the longitude and latitude of each city. Afterwards, I match the seafarer departures with recorded literacy rates by city and year<sup>1</sup>, allowing for an empirical analysis of their effects on the literacy gender gap.

To normalise the treatment variable in relation to a city’s initial size, I enrich the literacy dataset by adding Lourens and Lucassen’s (2025) digitised historical population estimates of Dutch cities, with pre-treatment population estimates being matched to the closest pre-treatment years before 1602. These pre-treatment counts are preferable to populations later on due to potential downward bias on the treatment effect, where a high number of seafarers may indicate higher levels of commercial trade and thereby a city’s economic prosperity, resulting in a larger population that drives the seafarer-population ratio downward. 22 cities have recorded population estimates around the year 1560, followed by 7 cities around 1600. Two of the three remaining cities differ slightly from the years around which most cities are concentrated to use more granular estimates, while Terschelling uses the 1632 population estimate, which is justifiable by a lack of population growth in between estimated periods (Lourens & Lucassen, 2025). Consequently, two cities, Texel and Vlieland, are dropped from the dataset due to missing population estimates<sup>2</sup>.

Figure 1 displays the geographical distribution of the 32 cities, showing that the sample is mostly evenly spread across the Dutch Republic. Most notably, Amsterdam’s literacy rates by gender are not present in the dataset, leaving out a key city which became exemplary for the Dutch Republic’s maritime golden age. Nevertheless, the sample still contains a sufficient representation of western cities such as Rotterdam and Leiden, which benefited from the VOC.

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<sup>1</sup>For more details on the enriched literacy dataset, see Appendix B.

<sup>2</sup>For more details on the city population data, see Appendix C.

**Figure 1**

*Geographical distribution of Dutch cities in the literacy dataset*



*Note.* Yellow areas roughly represent territories controlled by the Spanish Empire throughout the Eighty Years' War. Territorial boundaries are taken from Stapel (2025).

### 3.4 Summary statistics and balance tests

Table 1 shows summary statistics of 32 Dutch cities in the literacy dataset at 1600, 1675 and 1750. These statistics<sup>3</sup> confirm that Dutch cities start with a substantial literacy gender gap, which narrows over time primarily due to a mean increase in female literacy and a decrease in its variance across the 32 cities, outpacing the concurrent increase in male literacy. Most notably, a large variance exists in the observed city populations, which is initially corrected for by the normalised treatment variable.

Figure 2 displays the distribution of the treatment variable, showing that few of the 32 cities can be considered true control units or as good as untreated. As the VOC enlisted individuals all across the Dutch Republic (Petram et al., 2024), it is unsurprising that all cities have some level of seafarer departures over a century-long period, explaining why binary distinctions between untreated and treated groups of cities are infeasible. A notable outlier is Rotterdam, which causes a right-skewed treatment distribution and makes Rotterdam a large driver of the estimated effect in Section 4, which is addressed later on in robustness checks. Overall, the treatment variable has substantial variation across intensities, allowing for comparisons between cities with lower and higher treatment intensity.

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<sup>3</sup>The mean population size drops for Dutch cities in 1750, which cannot be explained by some small cities dropping out of the sample due to missing data. Leiden seems to be the primary driver in this regard, losing roughly half of their population as a result of a coinciding decline of its textile industry beginning from the 1670s (de Vries & van der Woude, 1997, pp. 289–290).

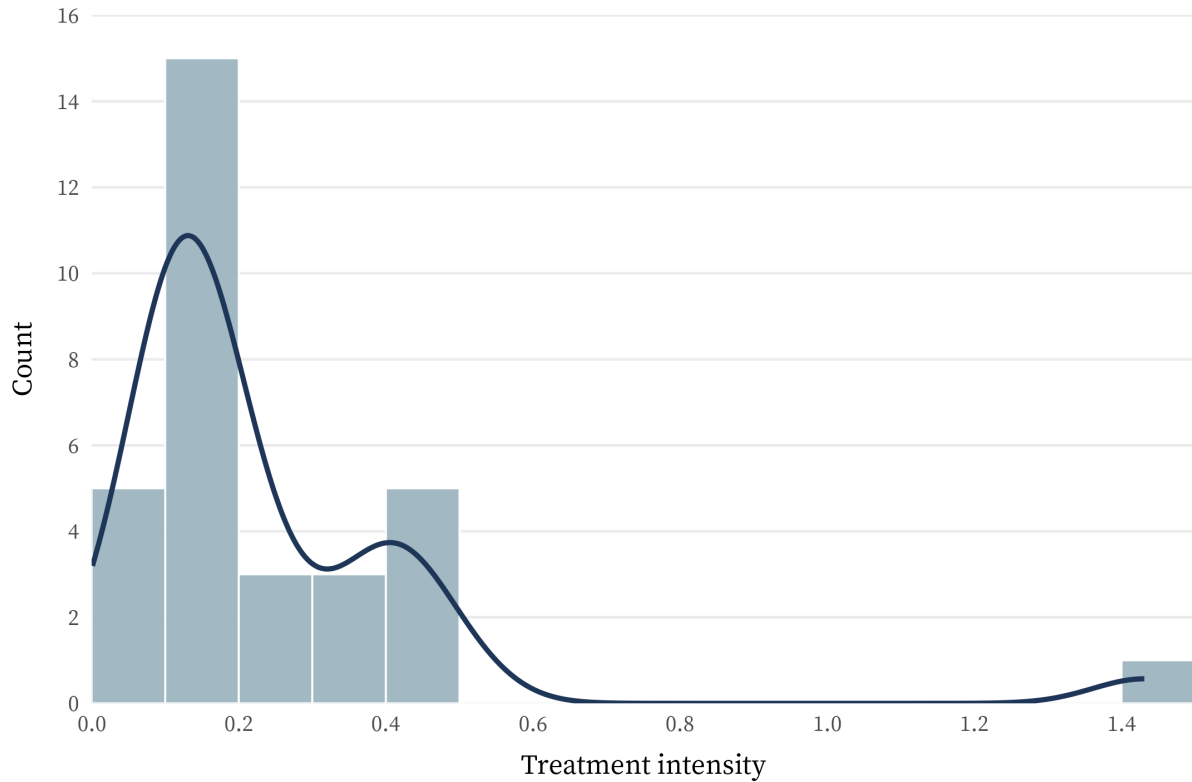
**Table 1***Summary statistics of 32 Dutch cities in 1600, 1675 and 1750*

| Variable            | Statistic | 1600      | 1675      | 1750      |
|---------------------|-----------|-----------|-----------|-----------|
| Female literacy     | N         | 32        | 32        | 32        |
|                     | Mean      | 33.45     | 48.08     | 59.79     |
|                     | SD        | 12.17     | 10.15     | 9.75      |
|                     | Min       | 9.74      | 20.69     | 30.77     |
|                     | Max       | 57.50     | 66.07     | 75.00     |
| Male literacy       | N         | 32        | 32        | 32        |
|                     | Mean      | 66.63     | 75.86     | 85.36     |
|                     | SD        | 9.38      | 8.89      | 6.42      |
|                     | Min       | 44.36     | 50.62     | 71.43     |
|                     | Max       | 84.21     | 90.82     | 95.92     |
| Literacy gender gap | N         | 32        | 32        | 32        |
|                     | Mean      | -33.19    | -27.78    | -25.57    |
|                     | SD        | 10.02     | 10.95     | 9.32      |
|                     | Min       | -54.77    | -58.80    | -52.56    |
|                     | Max       | -10.04    | -10.19    | -10.03    |
| Population          | N         | 32        | 31        | 29        |
|                     | Mean      | 6,011.94  | 12,135.35 | 11,613.97 |
|                     | SD        | 5,152.62  | 14,669.86 | 12,786.47 |
|                     | Min       | 467.00    | 600.00    | 1,200.00  |
|                     | Max       | 17,500.00 | 67,000.00 | 56,000.00 |
| Treatment intensity | N         | 32        | 32        | 32        |
|                     | Mean      | 0.24      | 0.24      | 0.24      |
|                     | SD        | 0.25      | 0.25      | 0.25      |
|                     | Min       | 0.00      | 0.00      | 0.00      |
|                     | Max       | 1.43      | 1.43      | 1.43      |
| Longitude           | N         | 32        | 32        | 32        |
|                     | Mean      | 5.60      | 5.60      | 5.60      |
|                     | SD        | 0.77      | 0.77      | 0.77      |
|                     | Min       | 4.30      | 4.30      | 4.30      |
|                     | Max       | 6.93      | 6.93      | 6.93      |
| Latitude            | N         | 32        | 32        | 32        |
|                     | Mean      | 52.42     | 52.42     | 52.42     |
|                     | SD        | 0.46      | 0.46      | 0.46      |
|                     | Min       | 51.81     | 51.81     | 51.81     |
|                     | Max       | 53.36     | 53.36     | 53.36     |

Table rows report observations, mean, standard deviation, minimum and maximum by year. Literacy rates are in percentages. Population values are recorded in inhabitants; N can differ by period when cities have missing population data in 1675 and/or 1750. Treatment intensity is time-invariant for the purpose of a difference-in-differences analysis.

**Figure 2**

*Treatment intensity distribution across 32 Dutch cities*



*Note.* The smooth curve represents a density curve fitted onto the distribution, with its total area equal to 1. A bin width of 0.1 is used.

Imbalanced pre-treatment city characteristics may bias the estimated causal effect of seafarer departures on the literacy gender gap. If city characteristics significantly vary depending on the treatment intensity, the interpretation of the estimated causal effect may be confounded, and the parallel trends assumption underlying the empirical approach mentioned in Section 4 may not hold. For example, if cities with larger populations are significantly associated with receiving a higher treatment intensity, seafarer departures are then not distributed as-good-as-random across cities. The treatment is then concentrated in cities with stronger population growth trends, which proxy for higher economic development trajectories (Cantoni, 2015, pp. 569–570).

Consequently, cities on higher economic development trajectories may have urban institutions that are more conducive to women acquiring literacy to benefit the maritime economy, especially in cities with access to the booming Atlantic trade (Acemoglu et al., 2005). The rise of cities in the Dutch Republic such as Rotterdam coincided with the rise of their merchant groups to political power, allowing them to instill urban cultural values that, driven by commercial self-interest, were more likely to be supportive of

women than those of the nobility and the clergy (de Vries & van der Woude, 1997, pp. 164–165). Thus, such a positive statistical association can violate the parallel trends assumption without properly accounting for population growth trends that feed back into the literacy gender gap.

To test for such imbalances, I run a series of balance tests by regressing the treatment variable on pre-treatment city characteristics listed in Table 1. Table 2 shows that the treatment variable is initially not significantly associated with pre-treatment city characteristics, but the coefficient for city population size is directionally problematic: thus, I log transform the population size to account for a non-linear relationship between treatment intensity and population size, resulting in a significant and positive association between treatment intensity and log-transformed population size at the 10% level. Additionally, treatment intensity is negatively associated with longitude, which suggests that closeness to the western coast and thereby the Atlantic trade is associated with higher treatment intensity.

Given the low number of observed cities, methods such as propensity score matching to obtain comparable cities are infeasible, and further pre-trend tests with regard to the parallel trends assumption are not possible due to the presence of only one period prior to treatment. As similar trajectories rather than similar levels matter for the parallel trends assumption and that growth trajectories better signal the rise in political prominence of the city’s urban elites, I control for available population growth trends in the periods 1600-1675 and 1675-1750 in the empirical strategy and control for longitude and latitude interacted with time to increase the plausibility of fulfilling the conditional parallel trends assumption by absorbing differential trends.

**Table 2**

*Pre-treatment covariate balance tests in 1600*

| Outcome             | Estimate | Std. Error | p-value |
|---------------------|----------|------------|---------|
| Female literacy     | -2.67    | 6.38       | 0.679   |
| Male literacy       | -6.27    | 6.02       | 0.306   |
| Literacy gender gap | 3.60     | 3.73       | 0.341   |
| Population          | 3643.02  | 2642.06    | 0.178   |
| ln(Population)      | 0.92     | 0.48       | 0.062   |
| Latitude            | -0.11    | 0.29       | 0.703   |
| Longitude           | -1.64    | 0.57       | 0.007   |

Each regression uses the 1600 cross-section only (N = 32) and regresses the listed outcome on the treatment intensity. Standard errors are heteroskedasticity-robust.

## 4 Maritime-induced male absence and the literacy gender gap

This section delves into identifying the causal effect of maritime-induced male absence, measured by seafarer departures, on the literacy gender gap through a continuous difference-in-differences (DiD) approach, accounting for the limitations and potential violations of its underlying assumptions mentioned in Section 3. Later on, robustness checks such as additional population controls and sensitivity tests are presented to test the validity of these results.

### 4.1 Empirical strategy

To estimate the causal effect of seafarer departures on the literacy gender gap, I use a DiD approach that exploits variation in the number of seafarer departures across the Dutch Republic. By comparing the changes in the literacy gender gap before and after the pre-treatment period of 1600 given the variation in seafarer departures over two treatment periods, a three-period DiD design allows for the identification of the average treatment effect (Roth et al., 2023).

In contrast to the canonical DiD design, I use a continuous treatment variable that allows for comparisons beyond binary distinctions between treated and untreated cities, which in this research design would have required an arbitrary threshold to distinguish between cities with high and low VOC activity. A continuous DiD analysis, therefore, compares across levels of treatment intensity, allowing for an analysis of the case where every city is at least minimally exposed to the VOC (Callaway et al., 2024). Since the empirical design compares the effects of lower and higher doses of treatment, the estimated causal effect, if treatment effects are monotonic in treatment intensity, is likely to be understated compared to an effect distinguishing between untreated and treated cities.

To make it more plausible that the conditional parallel trends assumption holds, which is key to correctly identifying the average treatment effect resulting from a DiD analysis, I include interactions of longitude and latitude with year fixed effects to control for differential trends across cities, while also controlling for population growth to account for the influence of maritime commercial development on literacy gender gap trends. In particular for the former, longitude proxies for closeness to the coast, where the western Dutch cities were commercially booming and had a higher propensity to have institutions which benefited commercial interests (Acemoglu et al., 2005). Simultaneously, urban institutions there could also be exclusionary to women, such as guilds. These organisations were generally not open for female membership, which hampered women from entering profitable trades (van den Heuvel, 2007, p. 138)



and likely discouraged them from acquiring literacy skills needed to operate in these trades.

For this approach, I use a two-way fixed effects (TWFE) estimator that controls for unobserved time-invariant characteristics of cities and common shocks over time. Under the conditional parallel trends assumption and the assumption that treatment effects are homogeneous across cities and time<sup>4</sup>, a TWFE estimator nominally yields the unbiased average treatment effect (Roth et al., 2023).

The following baseline two-way fixed effects regression is estimated:

$$y_{i,t}^f - y_{i,t}^m = \beta D_i \times Post_t + \gamma_i + \delta_t + \sum_{s \neq 1600}^{1750} \theta'_s X_{i,t} \times \mathbf{1}[t = s] + \varepsilon_{i,t}, \quad (1)$$

where  $y_{i,t}^f - y_{i,t}^m$  is the literacy gender gap in city  $i$  at year  $t$ , with  $t \in \{1600, 1675, 1750\}$ .  $D_i$  denotes the continuous treatment intensity, which is the aggregate number of seafarer departures from city  $i$ , relative to the city's pre-treatment population size.  $\beta$  is therefore the coefficient of interest, representing the post-treatment difference in the literacy gender gap. The pre-treatment population size is used to account for the effect of seafarer departures on city size growth, where, in some cities, this treatment may correlate with higher levels of commercial activity, leading to local institutions being more receptive to maritime traders (Acemoglu et al., 2005), who rationally had no interest in constraining the labour pool they could draw from by reinforcing gender norms around literacy. For preciseness,  $D_i$  is therefore constructed as follows:

$$D_i = \frac{\text{departures}_i}{\text{population}_{i,1600}}. \quad (2)$$

Moving on,  $Post_t$  is an indicator variable that equals 1 for the post-treatment years 1675 and 1750.  $\gamma_i$  and  $\delta_t$  represent city and year fixed effects, which absorb what would normally be the intercept  $\beta_0$ ,  $Post_t$  and the time-invariant  $D_i$  outside the post-treatment interaction term. Next,  $X'_{i,t}$  is a set of control variables including time-invariant longitude, latitude and time-varying population growth, which are interacted with year fixed effects to absorb the differential trends of cities in relation to these time-invariant and time-variant city characteristics. The inclusion of longitude and latitude as control variables follows from literature using similar identification strategies to estimate the

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<sup>4</sup>Callaway and Sant'Anna (2021) address how TWFE estimators deliver biased estimates when treatment timing is staggered and effects are heterogeneous across groups or time. This effect bias results from forbidden comparisons between already treated and newly treated units. In the case of the VOC, some cities such as those located in the east of the Dutch Republic were recorded years later as having seafarer departures compared to western cities. However, as there is only outcome data available in 75-year intervals from 1600 to 1750, mapping staggered adoption leads to nearly uniform adoption in practice. Given more detailed literacy data, the use of a heterogeneity-robust estimator as proposed by Callaway et al. (2024) would have been a necessary robustness check.

effect of historical shocks (Cantoni, 2015; Jia, 2014). To avoid perfect multicollinearity, 1600 as the pre-treatment year is chosen as the baseline and omitted as an interaction term.  $\mathbf{1}[t = s]$  is an indicator variable that equals 1 for 1675 and 1750. Finally,  $\varepsilon_{i,t}$  is the error term.

As observations within cities over time are not independent, I also cluster standard errors at the city level to correct for error terms that are correlated within cities (Roth et al., 2023).

Treatment effects may be heterogeneous over time, and, in the case of maritime-induced male absence, take longer to fully unfold, as human capital accumulation decisions by women are unlikely to immediately change at the time of treatment. Therefore, I also estimate a flexible specification by interacting the treatment intensity  $D_i$  with the years 1600, 1675 and 1750. This gives the following equation:

$$y_{i,t}^f - y_{i,t}^m = \sum_{s \neq 1600}^{1750} \beta_t (D_i \times \mathbf{1}[t = s]) + \gamma_i + \delta_t + \sum_{s \neq 1600}^{1750} \theta'_t \mathbf{X}_{i,t} \times \mathbf{1}[t = s] + \varepsilon_{i,t}, \quad (3)$$

where 1600 is also chosen for the treatment baseline.

## 4.2 Results

Table 3 reports both the results of the baseline and flexible TWFE regression. Column (1) initially presents a puzzling estimate, with a negative effect of seafarer departures on the literacy gender gap. However, after adding longitude and latitude interactions with year fixed effects, column (2) and (3) show that longitude primarily flips the effect sign and reduces the standard error size, showing that the baseline estimate with no controls is biased downward by western cities experiencing widening gender gap trends that are unrelated to the treatment.

As mentioned previously, longitude can proxy for the distance to the coast, suggesting that coastal cities due to their maritime commercial development were on differential gender gap trends relative to eastern cities. Intuitively, this can also be explained in the other direction for eastern cities, which in the sample coincidentally includes many Hanseatic cities like Zwolle, Kampen and Deventer, which experienced their trade boom centuries earlier. In particular, Deventer was the origin place of the Brethren of the Common Life (BCL) in the 14th century, which has been estimated to be positively associated with higher literacy rates for both Dutch women and men long after the BCL had declined into irrelevance (Akçomak et al., 2016).

**Table 3***Effects of seafarer departures on the literacy gender gap*

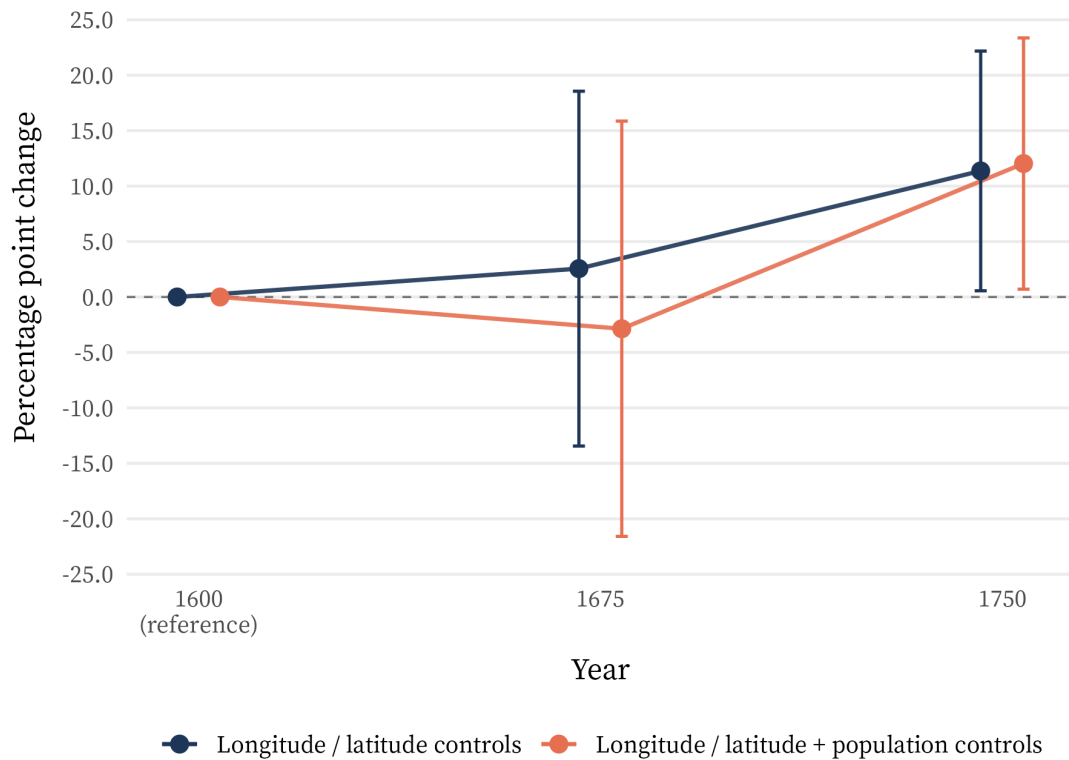
| Dependent variable            | Literacy gender gap |                |                |                 |                 |                   |                   |                   |
|-------------------------------|---------------------|----------------|----------------|-----------------|-----------------|-------------------|-------------------|-------------------|
|                               | (1)                 | (2)            | (3)            | (4)             | (5)             | (6)               | (7)               | (8)               |
| <i>Baseline specification</i> |                     |                |                |                 |                 |                   |                   |                   |
| $D_i \times Post_t$           | -3.27<br>(5.75)     | 6.81<br>(6.21) | 6.96<br>(5.76) | 6.81<br>(5.65)  |                 |                   |                   |                   |
| pop. growth $\times$ 1675     |                     |                |                | 1.28<br>(4.49)  |                 |                   |                   |                   |
| pop. growth $\times$ 1750     |                     |                |                | -6.47<br>(5.21) |                 |                   |                   |                   |
| <i>Flexible specification</i> |                     |                |                |                 |                 |                   |                   |                   |
| $D_i \times 1675$             |                     |                |                |                 | -8.64<br>(7.68) | 2.37<br>(8.77)    | 2.56<br>(7.84)    | -2.86<br>(9.17)   |
| $D_i \times 1750$             |                     |                |                |                 | 2.11<br>(5.08)  | 11.26**<br>(5.25) | 11.37**<br>(5.30) | 12.03**<br>(5.55) |
| pop. growth $\times$ 1675     |                     |                |                |                 |                 |                   |                   | 6.18<br>(4.57)    |
| pop. growth $\times$ 1750     |                     |                |                |                 |                 |                   |                   | -8.62<br>(5.51)   |
| Longitude $\times$ Year FE    |                     | ✓              | ✓              | ✓               |                 | ✓                 | ✓                 | ✓                 |
| Latitude $\times$ Year FE     |                     |                | ✓              | ✓               |                 |                   | ✓                 | ✓                 |
| Population growth             |                     |                |                | ✓               |                 |                   |                   | ✓                 |
| Year FE                       | ✓                   | ✓              | ✓              | ✓               | ✓               | ✓                 | ✓                 | ✓                 |
| City FE                       | ✓                   | ✓              | ✓              | ✓               | ✓               | ✓                 | ✓                 | ✓                 |
| N                             | 96                  | 96             | 96             | 91              | 96              | 96                | 96                | 91                |
| R <sup>2</sup>                | 0.473               | 0.510          | 0.533          | 0.609           | 0.484           | 0.516             | 0.538             | 0.620             |
| Adj. R <sup>2</sup>           | 0.179               | 0.212          | 0.222          | 0.309           | 0.183           | 0.207             | 0.217             | 0.315             |
| Within R <sup>2</sup>         | 0.003               | 0.074          | 0.116          | 0.146           | 0.023           | 0.084             | 0.126             | 0.170             |

Clustered standard errors (city-level) are shown in parentheses.  $D_i$  denotes seafarer departures divided by baseline population. Coefficients represent percentage-point changes in the dependent variable. All specifications include city and year fixed effects. Population growth coefficients are reported where included. The sample contains 32 cities observed over 3 periods. The longitude and latitude interactions are reported in Table A1. Terschelling is dropped from specifications with population growth controls due to missing data, alongside Borculo and Groenlo's 1750 observations. Longitude and latitude interactions are reported in Table A1. Significance levels: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

The flexible specification in column (6) and (7) shows that when the effects of seafarer departures are able to vary over treatment periods, the estimated effect in 1750 is more pronounced compared to the baseline and statistically significant at the 5% level while controlling for longitude and latitude. Taken from column (7), this estimate suggests that a city in which a 25% equivalent of the 1600 baseline population flowed out to sea, equal to a one standard deviation increase in the treatment, experienced a 2.84

**Figure 3**

*Event study plot of the literacy gender gap*



*Note.* Point estimates with 95% confidence intervals of flexible specification (7) and (8) from Table 3 are displayed.

percentage point decrease<sup>5</sup> in the literacy gender gap. Figure 3 plots the point estimates over time in an event study plot.

Table 4 indicates the composition of the literacy gender gap estimate from column (7) of Table 3. The female literacy coefficient is statistically significant in both treatment periods respectively at the 10% and 5% level, accounting for 99.4% of the treatment effect on the gender gap in 1750. Male literacy coefficients are not statistically significant in any treatment period, suggesting that the effect of seafarer departures on literacy is likely to be solely applicable to Dutch women, confirming the conceptual framework behind the treatment. Altogether, these separate results signal that female literacy is the sole significant driver behind the closing of literacy gender gap.

<sup>5</sup>While the coefficients are positively signed, the context behind the literacy gender gap as a form of inequality and taking into account its construction as female minus male literacy leads me to refer to positive coefficients as a decrease or closing of the literacy gender gap.

**Table 4***Effects of seafarer departures on female and male literacy*

| Dependent variable         | Female literacy |         | Male literacy |        |
|----------------------------|-----------------|---------|---------------|--------|
|                            | (1)             | (2)     | (3)           | (4)    |
| $D_i \times 1675$          | 8.10*           | 6.97    | 5.54          | 9.93   |
|                            | (4.66)          | (6.72)  | (5.89)        | (7.47) |
| $D_i \times 1750$          | 11.30**         | 14.25** | -0.07         | 2.16   |
|                            | (4.13)          | (5.85)  | (3.44)        | (3.23) |
| Longitude $\times$ Year FE | ✓               | ✓       | ✓             | ✓      |
| Latitude $\times$ Year FE  | ✓               | ✓       | ✓             | ✓      |
| Population growth          |                 | ✓       |               | ✓      |
| Year FE                    | ✓               | ✓       | ✓             | ✓      |
| City FE                    | ✓               | ✓       | ✓             | ✓      |
| N                          | 96              | 91      | 96            | 91     |
| R <sup>2</sup>             | 0.866           | 0.877   | 0.820         | 0.844  |
| Adj. R <sup>2</sup>        | 0.773           | 0.779   | 0.695         | 0.720  |
| Within R <sup>2</sup>      | 0.162           | 0.202   | 0.235         | 0.245  |

Clustered standard errors (city-level) are shown in parentheses.  $D_i$  denotes seafarer departures divided by baseline population. Coefficients represent percentage-point changes in the dependent variable. All specifications include city and year fixed effects. Significance levels: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

### 4.3 Robustness checks

To test the sensitivity of the statistically significant treatment effects, I perform two robustness checks using post-treatment population growth as a control variable and dropping the largest driver of the results, namely Rotterdam, which is an outlier in terms of treatment intensity.

Column (8) of Table 3 shows that the treatment effect in 1750 minimally changes in magnitude and remains statistically significant at the 5% level after controlling for population growth. Population growth as its own coefficient produces an insignificant coefficient, indicating that population growth trends are likely less relevant for explaining the closing of the literacy gender gap. The results of this robustness check suggests that population growth trends by year may not meaningfully confound the estimated treatment effect.

A caveat to any stronger conclusion of this robustness check is that no consistent statistics exist to construct pre-treatment population growth trends, which precludes a pre-post comparison of growth trends that can help distinguish between pre-existing trajectories and maritime-induced growth. Pre-trends would be necessary to both

contribute to the plausibility of the conditional parallel trends assumption and establish whether population growth trends, reflecting the economic development trajectory of cities at that time, do not confound the treatment effect.

Next, Table 5 shows the adjusted treatment effects after leaving out Rotterdam, a major outlier in the treatment intensity distribution. All treatment effects become statistically insignificant with far larger standard errors, confirming that the estimated treatment effects are highly driven by Rotterdam.

The established sensitivity of these coefficients to dropping the most treated city, however, does not necessarily invalidate the previous results. Removing a prominent VOC chamber city in the form of Rotterdam is bound to extinguish a large proportion of the average treatment effect, and unavailable literacy data for Amsterdam and Middelburg, two key VOC chamber cities, further diminishes a possibly meaningful conclusion from the leave-one-out test. Nonetheless, this robustness check still does show that a large city such as Rotterdam, a large founding city of the VOC, likely experienced larger effects of seafarer departures on the literacy gender gap than other cities.

**Table 5***Leave-one-out: effects excluding Rotterdam*

| Dependent variable            | Literacy gender gap |                  |
|-------------------------------|---------------------|------------------|
|                               | (1)                 | (2)              |
| <i>Flexible specification</i> |                     |                  |
| $D_i \times 1675$             | 1.98<br>(27.04)     | -2.81<br>(31.73) |
| $D_i \times 1750$             | 20.31<br>(19.01)    | 6.82<br>(13.57)  |
| pop. growth $\times$ 1675     |                     | 5.98<br>(5.07)   |
| pop. growth $\times$ 1750     |                     | -9.02<br>(5.29)  |
| Longitude $\times$ Year FE    | ✓                   | ✓                |
| Latitude $\times$ Year FE     | ✓                   | ✓                |
| Population growth             |                     | ✓                |
| Year FE                       | ✓                   | ✓                |
| City FE                       | ✓                   | ✓                |
| N                             | 93                  | 88               |
| R <sup>2</sup>                | 0.532               | 0.612            |
| Adj. R <sup>2</sup>           | 0.203               | 0.298            |
| Within R <sup>2</sup>         | 0.124               | 0.158            |

Clustered standard errors (city-level) are shown in parentheses.  $D_i$  denotes seafarer departures divided by baseline population. Coefficients represent percentage-point changes in the dependent variable. Specifications exclude Rotterdam (location\_id = 34). All specifications include city and year fixed effects. Population growth coefficients are reported where included. The sample contains 31 cities observed over 3 periods after excluding Rotterdam. Significance levels: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## 5 Discussion and conclusion

This thesis finds through a difference-in-differences analysis using city-level literacy data by gender that VOC seafarer departures have a significant closing effect on the literacy gender gap. Cities in 1750 with a one standard deviation increase in seafarer departures relative to their city population size in 1600 experience an estimated 2.84 percentage point decrease in the literacy gender gap after controlling for longitude. This effect is almost entirely driven by increases in female literacy rather than decreases in male literacy. This finding is also robust to controlling for post-treatment population

growth trends, but far less robust to dropping the outlier city of Rotterdam, which due to a lack of data availability gives little generalisable conclusions about the true strength of the average treatment effect.

These results are consistent with the broader theme of this thesis, namely whether women's economic autonomy in the Dutch Republic was positively affected by maritime-induced male absence. The findings provide suggestive evidence in favour of a salient shock to the adult sex ratio, which divulged into consequences for domestic marriage markets and labour markets that encouraged women to acquire human capital.

Multiple valuable contributions emerge from this thesis. Importantly, treatment effects are found to only become statistically significant in 1750, signalling that seafarer departures have a prolonged and a long-run rather than short-run effect on the literacy gender gap. This finding provides a valuable contribution to the literature by showing that long-lasting historical shocks such as maritime-induced male absence can have persistent effects on women's economic outcomes. Generally, this contribution applies more to shocks affecting relatively marginalised rather than already empowered groups of the population. Additionally, this thesis contributes to the study of historical shocks with in the background the Protestant Reformation and other religious movements before it, which have been shown to have positive effects on female outcomes (Akçomak et al., 2016; Becker & Woessmann, 2008). Finally, this thesis fills the gap in the quantitative literature that presents explanations for why women in the Dutch Republic seemed economically more independent than in other countries (Schmidt, 2025).

A limitation of this thesis concerning external validity follows from its second contribution in relation to the Protestant Reformation. As this thesis studies the Dutch Republic, and the data only contained Dutch cities as valid units of analysis, the thesis' contribution does not show whether the effects of seafarer departures are robust to variation in the religious status of cities, where cities in the broader Netherlands region were involved in the Eighty Years' War that had put religious denomination at its core. This variation was largely absent in the Dutch Republic, as its territories were nominally Protestant. With available literacy data on modern-day Belgian cities, further tests would be required to show whether the effect of the historical shock is enabled by the Protestant context of the Dutch Republic, or if the effect is also extendable to other religious environments.

Internal limitations of this thesis include the small sample size of 32 Dutch cities and the lack of pre-treatment periods, making pre-trend tests impossible and making it difficult to argue that the parallel trends assumption is not violated. Support for the conditional parallel trends assumption is hard to establish: little historical data from the early modern period exists on more granular economic variables that can be included in research to control for covariates in both pre- and post-treatment periods. Yet, the thesis attempts to establish support nonetheless by controlling for differential pre-treatment



and post-treatment longitude and latitude trends and post-treatment population growth trends. Additionally, with more available time periods associated with literacy rates and seafarer departures, heterogeneity-robust DiD estimators as proposed by Callaway et al. (2024) would be required to assess whether the thesis' findings remain robust.

Furthermore, the thesis' estimated treatment effects on the one hand could be seen as an upper bound given the statistically insignificant results when dropping the outlier city of Rotterdam. On the other hand, arguments could exist for the coefficients being a lower bound of the average treatment effect, as no true control units exist in the Dutch case. Comparing lightly treated units with heavily treated units, therefore, is likely to obtain an estimate that is lower than with a clean comparison between control and treatment groups.

This thesis also opens up normative discussions on colonialism and marginalised groups. Logically and predominantly, the discourse on European colonialism centers around the obvious group of oppressed countries and their people in Africa, the Americas and Asia. For example, Fanon (1961)'s well-known philosophical work on decolonisation delves into the eviction of colonial violence through sometimes equally violent means. Little focus is then given to marginalised groups within the colonisers, where this thesis shows in the case of the Dutch Republic that women could elevate their economic autonomy as a result of men being pulled away by a company heavily involved with extractive activities outside Europe. Marginalised groups in the age of colonialism do not merely exist in colonised areas, but also exist in colonising states. Thus, the thesis' findings open up interesting discussions that shift the traditional focus of discourse on colonialism: the normative perception of these results is dependent on how disjointed the domestic effects of a colonial-linked shock should be analysed from the oppressive effects of colonialism abroad. Positive effects for the coloniser's marginalised groups may outweigh or, more likely, be outweighed by negative effects for the colonised.

Despite the historical setting of this thesis, the findings have contemporary implications for society. First, historical shocks studied in the thesis' strand of literature have been shown to have persistent effects, even on cultural traits like gender norms which are sometimes considered as immalleable: this thesis in particular shows how, despite the decline of the VOC in the 18th century (de Vries & van der Woude, 1997, pp. 448–449), seafarer departures then started manifesting as a positive long-run effect on the literacy gender gap. Historical shocks can be equivalent to policy interventions: large-sweeping policy changes beneficial for women by, for example, affecting labour and marriage markets, can improve their economic outcomes. The study of historical shocks provides a method through which historical events, whether short- or long-lasting, can offer lessons for today's policy.

Overall, this thesis provides valuable contributions to the economic history liter-

ature, implications for normative discussions beyond the scope of what is normally considered in the economic science and indirect insights for contemporary policy design. Through the study of a historical setting, this thesis links itself to important philosophical and policy debates today.

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## A Additional regression tables

**Table A1**

*Longitude and latitude coefficients in the literacy gender gap models*

| Covariate        | Baseline         | Baseline + pop. controls | Flexible         | Flexible + pop. controls |
|------------------|------------------|--------------------------|------------------|--------------------------|
| Longitude × 1675 | 7.49**<br>(3.14) | 7.98**<br>(3.54)         | 6.69*<br>(3.50)  | 8.32**<br>(3.46)         |
| Longitude × 1750 | 4.78*<br>(2.69)  | 6.69***<br>(2.37)        | 5.57**<br>(2.61) | 8.15***<br>(2.41)        |
| Latitude × 1675  | -8.49*<br>(4.79) | -5.75<br>(5.25)          | -8.46*<br>(4.84) | -7.63<br>(5.54)          |
| Latitude × 1750  | -4.64<br>(6.06)  | -9.08*<br>(4.96)         | -4.66<br>(6.05)  | -9.55*<br>(4.89)         |

These coefficients correspond to the longitude and latitude interactions included in the TWFE specifications for the literacy gender gap, shown with and without the population-growth controls. Coefficients represent level differences in relation to time-invariant longitude and latitude attributes by year. Standard errors are clustered at the city level. Significance levels: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## B Codebook for the enriched literacy dataset

Variables in the merged city-level literacy dataset used in the empirical analysis are summarised in Table B1. The table below records the variables used in the analysis, bar the variable province.

**Table B1**

*Codebook for the female literacy and recruitment intensity dataset*

| Variable                                                 | Type      | Description                                              |
|----------------------------------------------------------|-----------|----------------------------------------------------------|
| <i>City names and literacy rates</i>                     |           |                                                          |
| province                                                 | character | Province or region label from the source literacy data.  |
| city                                                     | character | City name.                                               |
| m160025                                                  | numeric   | Male literacy rate, 1600–1625 (percent).                 |
| m1675                                                    | numeric   | Male literacy rate, 1675 (percent).                      |
| m1750                                                    | numeric   | Male literacy rate, 1750 (percent).                      |
| f160025                                                  | numeric   | Female literacy rate, 1600–1625 (percent).               |
| f1675                                                    | numeric   | Female literacy rate, 1675 (percent).                    |
| f1750                                                    | numeric   | Female literacy rate, 1750 (percent).                    |
| <i>VOC exposure, population statistics and geography</i> |           |                                                          |
| seafarers_16001750                                       | numeric   | Total VOC seafarer departures from the city, 1600–1750.  |
| pop_year_1600                                            | numeric   | Population year used for the pre-treatment denominator.  |
| pop_value_1600                                           | numeric   | Population value used for the pre-treatment denominator. |
| pop_year_1675                                            | numeric   | Population year near 1675.                               |
| pop_value_1675                                           | numeric   | Population value near 1675.                              |
| pop_year_1750                                            | numeric   | Population year near 1750.                               |
| pop_value_1750                                           | numeric   | Population value near 1750.                              |
| treatment_intensity                                      | numeric   | $\text{seafarers\_16001750} / \text{pop\_value\_1600}$ . |
| latitude                                                 | numeric   | City latitude.                                           |
| longitude                                                | numeric   | City longitude.                                          |

Population fields remain NA when no acceptable historical match was available, as outlined in Appendix C.

## **C Population statistics**

Population statistics and overrides are documented here, which are used to construct treatment intensity variables in the empirical analysis. Population data for later periods (1675, 1750) are included for population growth calculations.

All population figures are sourced from Lourens and Lucassen (1997) and Lourens and Lucassen (2025). Where necessary, manual overrides based on the information in Lourens and Lucassen (1997) were applied to ensure consistency, such as retaining a more granular estimate or not using Napoleonic-era censuses as a comparison to pre-Napoleonic estimates.

**Table C1**

*Population estimates for 32 Dutch cities*

| Province                        | City         | pop. year 1600 | pop. value 1600 | pop. year 1675 | pop. value 1675 | pop. year 1750 | pop. value 1750 | Notes                                 |
|---------------------------------|--------------|----------------|-----------------|----------------|-----------------|----------------|-----------------|---------------------------------------|
| <i>Gelderland (9 cities)</i>    |              |                |                 |                |                 |                |                 |                                       |
|                                 | Groenlo      | 1541           | 1148            | 1670           | 1400            | -              | -               | Manual override 1675                  |
|                                 | Borculo      | 1560           | 600             | 1670           | 700             | -              | -               | Manual override 1675                  |
|                                 | Nijkerk      | 1560           | 1200            | 1650           | 1400            | 1749           | 2400            | Manual override 1600                  |
|                                 | Tiel         | 1560           | 2500            | 1650           | 2300            | 1741           | 3000            | Manual override 1750                  |
|                                 | Harderwijk   | 1560           | 2500            | 1650           | 3000            | 1749           | 2350            | Manual override 1600                  |
|                                 | Doesburg     | 1559           | 2093            | 1653           | 3266            | 1741           | 3000            | Manual override 1750                  |
|                                 | Arnhem       | 1600           | 7000            | 1670           | 6500            | 1749           | 5600            | Manual override 1675                  |
|                                 | Nijmegen     | 1560           | 10000           | 1670           | 9000            | 1741           | 12000           | Manual overrides 1675, 1750           |
|                                 | Zutphen      | 1560           | 6000            | 1672           | 5000            | 1741           | 5040            | Manual override 1750                  |
| <i>Noord-Holland (6 cities)</i> |              |                |                 |                |                 |                |                 |                                       |
|                                 | Alkmaar      | 1560           | 8000            | 1660           | 13650           | 1732           | 12500           | Manual override 1600                  |
|                                 | Haarlem      | 1560           | 16000           | 1665           | 38000           | 1748           | 24695           | Manual override 1600                  |
|                                 | Hoorn        | 1550           | 8000            | 1670           | 16000           | 1732           | 11060           | Manual override 1675                  |
|                                 | Weesp        | 1564           | 1700            | 1672           | 3300            | 1732           | 2400            | Auto-matched all                      |
|                                 | Enkhuizen    | 1561           | 7750            | 1632           | 19150           | 1732           | 9380            | Auto-matched 1600                     |
|                                 | Naarden      | 1564           | 1700            | 1670           | 2800            | 1732           | 2800            | Manual override 1675                  |
| <i>Overijssel (6 cities)</i>    |              |                |                 |                |                 |                |                 |                                       |
|                                 | Almelo       | 1560           | 800             | 1675           | 1170            | 1748           | 2229            | Manual overrides 1675, 1750           |
|                                 | Deventer     | 1599           | 6780            | 1675           | 6700            | 1748           | 7818            | Auto-matched 1600                     |
|                                 | Kampen       | 1599           | 6600            | 1675           | 5974            | 1748           | 5652            | Auto-matched 1600                     |
|                                 | Oldenzaal    | 1560           | 1500            | 1670           | 1600            | 1748           | 1960            | Manual override 1675                  |
|                                 | Ootmarsum    | 1560           | 600             | 1670           | 600             | 1748           | 1200            | Manual override 1675                  |
|                                 | Zwolle       | 1599           | 6500            | 1675           | 6963            | 1748           | 11931           | Auto-matched 1600                     |
| <i>Groningen (1 city)</i>       |              |                |                 |                |                 |                |                 |                                       |
|                                 | Groningen    | 1600           | 16600           | 1670           | 20600           | 1770           | 23296           | Manual override 1675                  |
| <i>Friesland (5 cities)</i>     |              |                |                 |                |                 |                |                 |                                       |
|                                 | Dokkum       | 1560           | 2000            | 1689           | 3177            | 1744           | 2855            | Manual override 1600                  |
|                                 | Franeker     | 1560           | 2200            | 1689           | 3623            | 1744           | 3493            | Manual override 1600                  |
|                                 | Harlingen    | 1560           | 2600            | 1689           | 8823            | 1744           | 7141            | Manual override 1600                  |
|                                 | Leeuwarden   | 1565           | 6600            | 1666           | 16500           | 1744           | 13490           | Auto-matched 1600                     |
|                                 | Terschelling | 1632           | 467             | -              | -               | -              | -               | Manual override 1600 (year exception) |
| <i>Zuid-Holland (5 cities)</i>  |              |                |                 |                |                 |                |                 |                                       |
|                                 | Delft        | 1600           | 17500           | 1680           | 24000           | 1749           | 13910           | Auto-matched 1600                     |
|                                 | Dordrecht    | 1560           | 15500           | 1665           | 20000           | 1732           | 18000           | Manual override 1600                  |
|                                 | s-Gravenhage | 1600           | 10000           | 1677           | 24000           | 1732           | 33500           | Manual overrides 1600, 1675, 1750     |
|                                 | Leiden       | 1581           | 12144           | 1665           | 67000           | 1750           | 38105           | Auto-matched 1600                     |
|                                 | Rotterdam    | 1561           | 7800            | 1665           | 40000           | 1732           | 56000           | Auto-matched 1600                     |

‘-’ denotes missing data or data excluded per the population matching rules. The Notes column indicates whether each city’s 1600 baseline was auto-matched from the population dataset or assigned a manual override. All 1600 baseline years satisfy the constraint year  $\leq 1602$ , except Terschelling which uses year 1632 by explicit exception.